Lean 4 Stokes Blueprint

May 24, 2026

Chapter 1

Stokes formalization blueprint

This blueprint is a dependency map for the Lean 4 Stokes project. The unit of work is a public definition, lemma, theorem, or a deliberately named future interface. Green nodes in the figures are already implemented in Lean; orange nodes are fieldized blockers whose constructors exist but whose geometric or measure-theoretic instances are still genuine work; blue nodes are open theorem targets intended to be small enough for independent assignment.

The current formalized stack already reaches boundary-chart local Stokes. The remaining work is not ”prove Stokes” as one block: it is the finite-cover, partition-of-unity, and global-integral assembly layer recorded in Section 1.4.

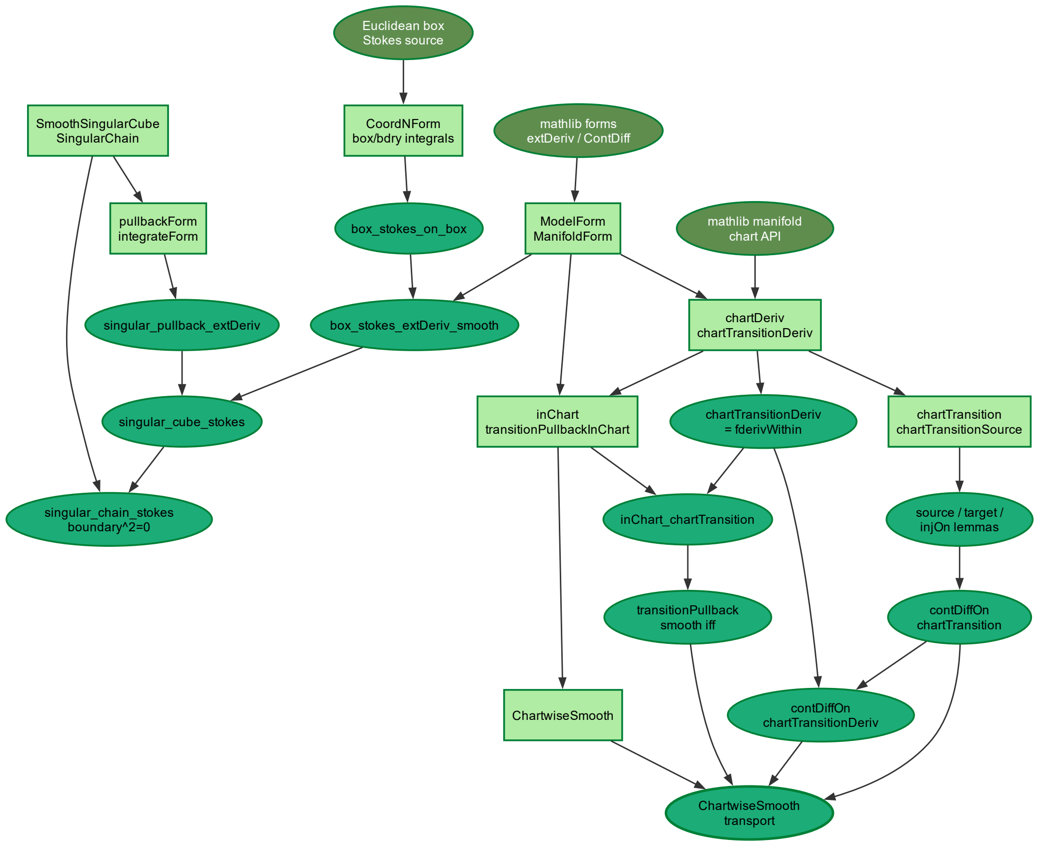

Figure 1.1: Core Euclidean, singular-cube, and chart-representative API.

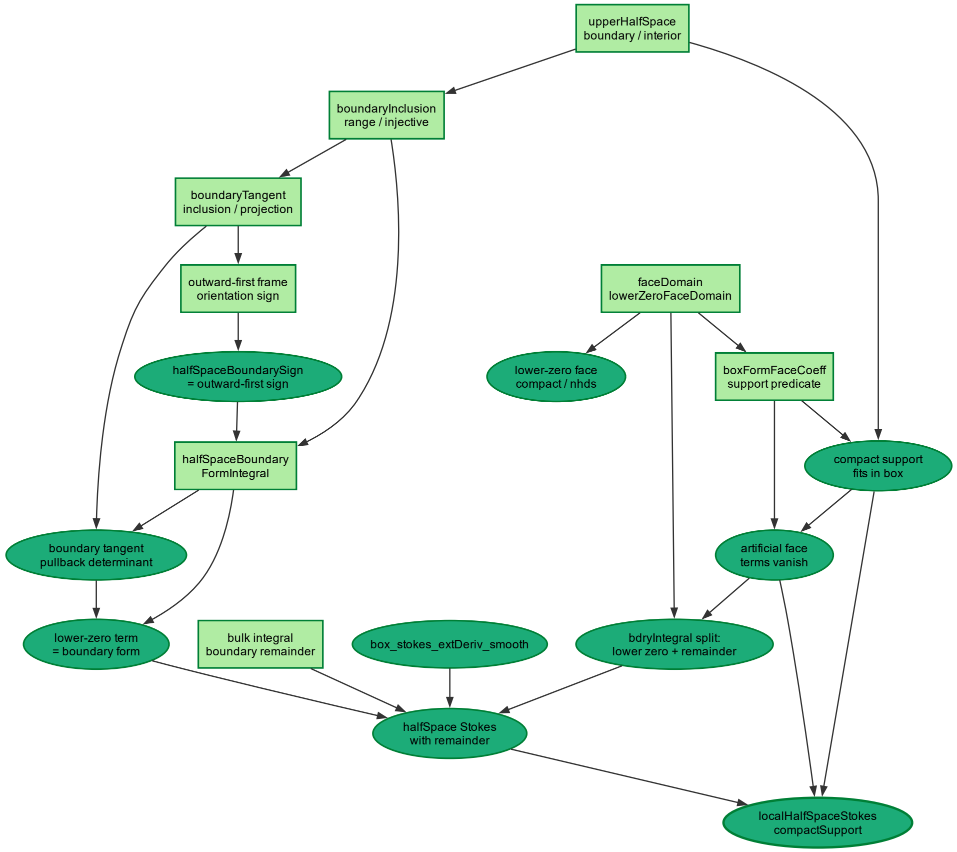

Figure 1.2: Half-space local Stokes and artificial-face cancellation.

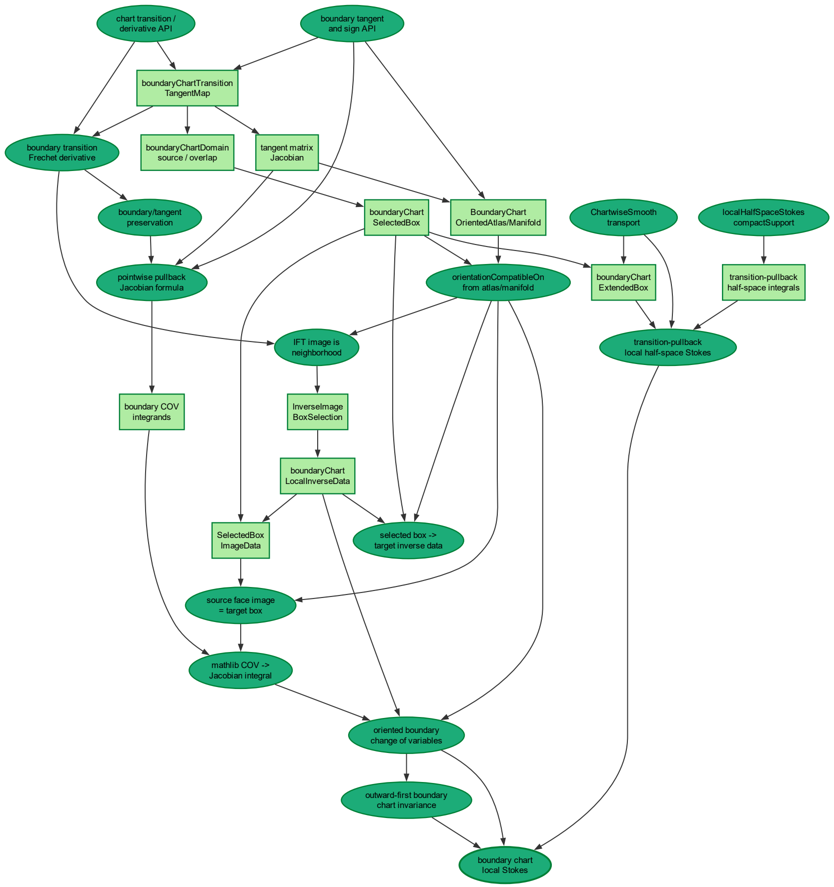

Figure 1.3: Boundary chart change of variables and local inverse data.

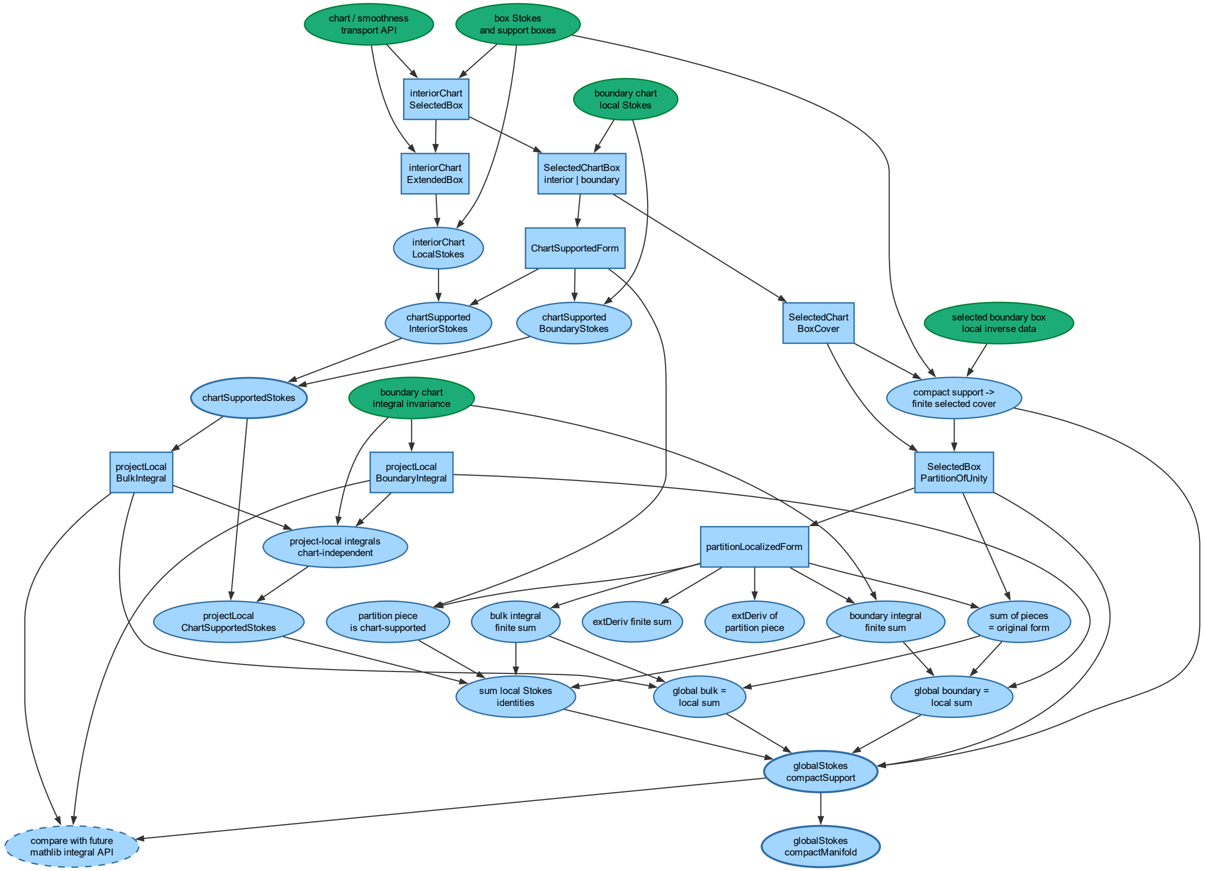

Figure 1.4: Remaining global assembly tasks.

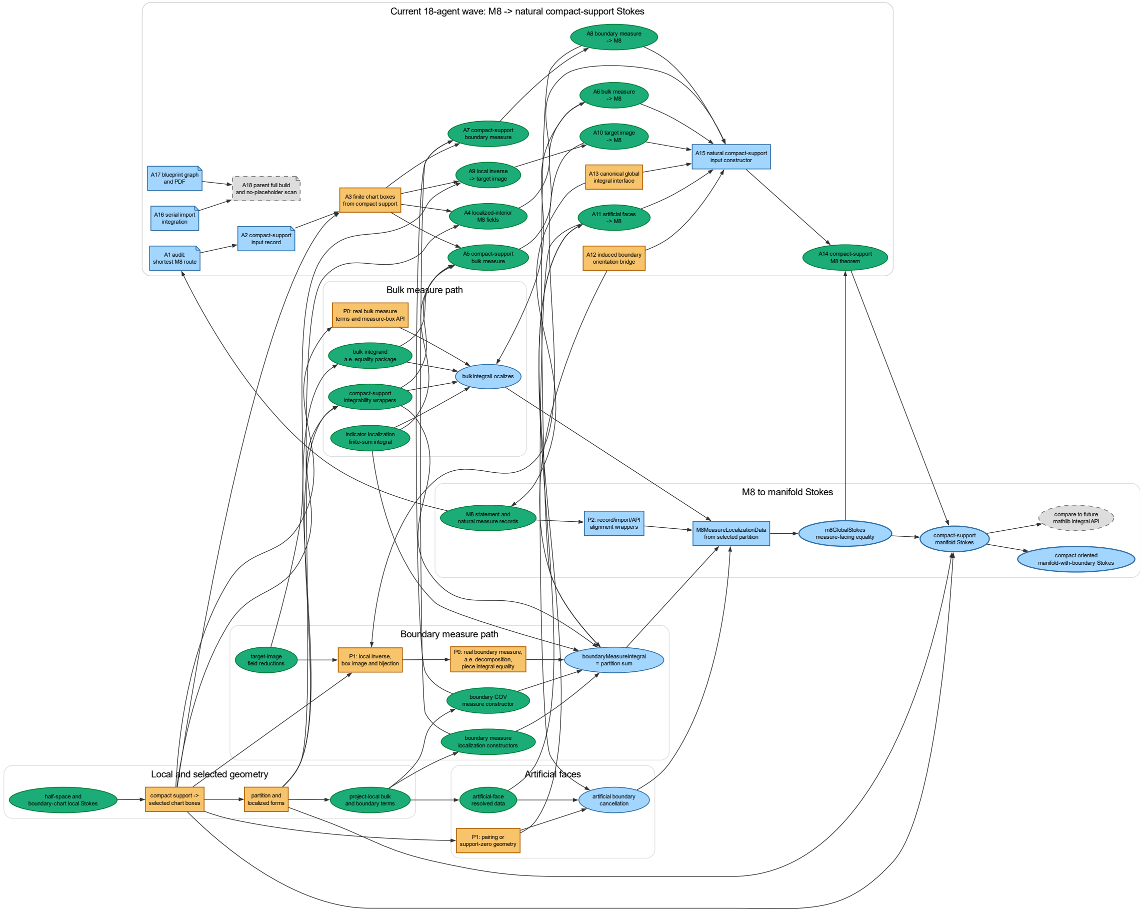

Figure 1.5: M8 measure localization, the compact-support wave, and route to manifold Stokes.

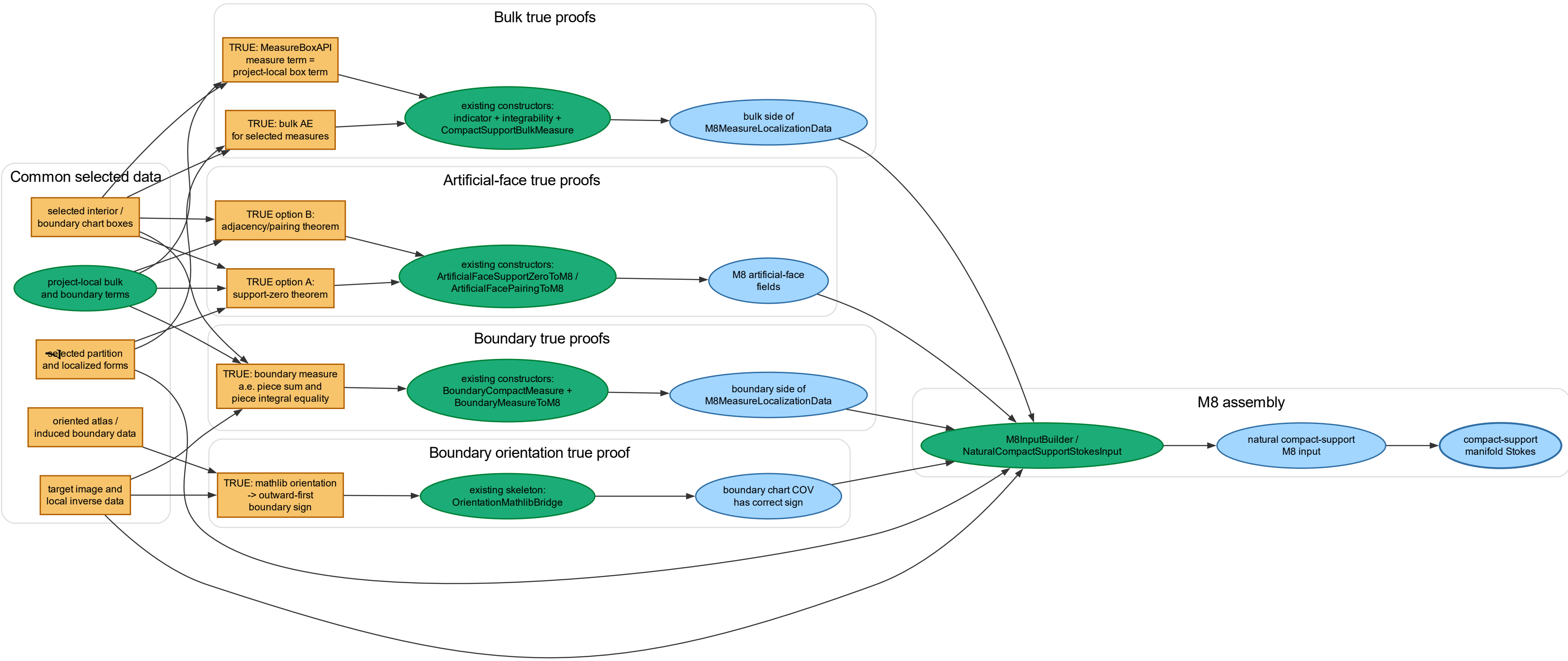

Figure 1.6: Current M8 proof wave: true theorem nodes behind the compact-support route.

1.1 Core Euclidean And Chart API

Definition 1.1 (Model and manifold forms).
A model form is a topological vector-space differential form in coordinates. A manifold form is a fiberwise alternating map on tangent spaces.

Definition 1.2 (Box integral interface).
This is the thin coordinate API connecting the project to the Euclidean box Stokes theorem supplied by the cloned local Euclidean stack.

Theorem 1.3 (Box Stokes on a rectangular box).
The signed boundary integral of a coordinate form on an axis-aligned box is the integral of its coordinate exterior derivative.

Theorem 1.4 (Box Stokes for mathlib forms).
The coordinate theorem is repackaged for mathlib-style differential forms and `extDeriv`.

Definition 1.5 (Smooth singular cubes and chains).
Smooth singular cubes, their cubical faces, finite chains, and chain integration provide an intermediate target between boxes and manifolds.

Definition 1.6 (Pullback form along a cube).
A smooth cube pulls back an ambient Euclidean form to a form on the standard cube, where the box theorem applies.

Theorem 1.7 (Exterior derivative commutes with cube pullback).
Pulling a form back along a smooth singular cube commutes with the exterior derivative in the shape needed by the cubical Stokes theorem.

Theorem 1.8 (Smooth singular cube Stokes).
Stokes holds for one smooth singular cube after expanding its oriented faces.

Theorem 1.9 (Smooth singular chain Stokes).
The cube statement extends linearly to finite chains, with boundary squared equal to zero.

Definition 1.10 (Chart derivatives and transitions).
The chart derivative API fixes the Frechet derivative maps used to express form pullback under coordinate changes.

Lemma 1.11 (Chart transition source and injectivity).
Chart transitions are defined on the expected overlap, map into the target chart, and are injective on their source.

Theorem 1.12 (Smoothness of chart transitions).
Mathlib's manifold chart API supplies the `ContDiffOn` facts needed to compose chart representatives with transition maps.

Theorem 1.13 (Transition derivative as Frechet derivative).
The concrete chart-transition derivative agrees with the Frechet derivative within the chart source.

Theorem 1.14 (Smoothness of transition derivatives).
The derivative-valued transition map is smooth on overlaps.

Definition 1.15 (Chart representative and transition pullback).
A manifold form is represented in a chart, and the same representative can be transported through a chart transition using the transition derivative.

Theorem 1.16 (Chart representative compatibility).
On a chart overlap, the chart representative is equal to the transition pullback written in the source coordinates.

Theorem 1.17 (Transition-pullback smoothness iff chart smoothness).
Smoothness of the transported coordinate form is equivalent to smoothness of the source chart representative on the chosen overlap.

Definition 1.18 (Chartwise smooth manifold form).
Chartwise smoothness packages the local smoothness hypotheses that later local Stokes statements consume.

Theorem 1.19 (Chartwise smooth transport).
A chartwise smooth form remains smooth after transporting it through a chart transition.

1.2 Half-Space Local Stokes

Definition 1.20 (Upper half-space model).
A concrete upper half-space in $\mathbb{R}^{n+1}$ fixes the local boundary model.

Definition 1.21 (Boundary inclusion and tangent maps).
The boundary hyperplane is identified with $\mathbb{R}^n$, together with the tangent inclusion and projection used by boundary forms.

Definition 1.22 (Outward-first frame and sign).
The lower face of the box has the same sign as the outward-normal-first boundary orientation.

Definition 1.23 (Box faces and face coefficients).
The box boundary integral is decomposed into the true boundary face and the artificial faces introduced by localizing to a box.

Lemma 1.24 (Compact support fits in a half-space box).
Compact support contained in the half-space can be enclosed in a box whose artificial faces miss the support.

Lemma 1.25 (Boundary face domain compactness).
A boundary face domain is compact and can be shrunk or selected inside a prescribed neighborhood.

Theorem 1.26 (Artificial face terms vanish).
If the form is supported in the selected half-space box, every artificial face contribution is zero.

Theorem 1.27 (Boundary integral splits into true face plus remainder).
The box boundary integral separates the lower-zero face from all remaining artificial faces.

Definition 1.28 (Half-space boundary integral).
The true boundary term is written as an integral over the boundary hyperplane using the outward-first sign convention.

Theorem 1.29 (Boundary tangent pullback determinant formula).
Pulling a top-degree boundary form through a linear tangent map multiplies the coordinate coefficient by the determinant.

Theorem 1.30 (Lower-zero face equals boundary form).
The lower-zero face coefficient extracted from the box is exactly the half-space boundary form coefficient.

Definition 1.31 (Half-space local bulk and remainder).
The local theorem is stated as a bulk integral equals a boundary integral plus a remainder from artificial faces.

Theorem 1.32 (Half-space Stokes with remainder).
The box theorem gives local half-space Stokes before canceling artificial faces.

Theorem 1.33 (Half-space Stokes with compact support).
Once support is contained in the selected half-space box, the remainder vanishes and the local half-space Stokes theorem is obtained.

1.3 Boundary Chart Change

Definition 1.34 (Boundary chart transition).
A boundary chart transition is the tangential coordinate map induced by an ambient chart transition preserving the boundary hyperplane.

Theorem 1.35 (Boundary chart transition derivative).
The Frechet derivative of the boundary chart transition is the tangential map obtained by restricting the ambient chart derivative.

Definition 1.36 (Boundary Jacobian).
The tangential derivative is converted to a matrix determinant for the Euclidean change-of-variables theorem.

Theorem 1.37 (Boundary preservation and tangent preservation).
The chart transition and its derivative preserve the boundary tangent hyperplane under the expected coordinate zero hypotheses.

Theorem 1.38 (Pointwise boundary pullback determinant).
The boundary integrand after chart change is the original boundary integrand multiplied by the tangential Jacobian.

Definition 1.39 (Boundary chart domain).
The boundary transition source is related to the ambient chart-transition source and overlap.

Definition 1.40 (Selected boundary chart box).
A selected source box records the local domain, support containment, and boundary-face overlap facts needed by local Stokes.

Definition 1.41 (Extended boundary chart box).
The selected box is strengthened with a neighborhood where the transported chart form has `ContDiffOn` smoothness.

Definition 1.42 (Boundary chart oriented atlas).
This is the local orientation bridge: oriented boundary charts have positive tangential Jacobian on the relevant boundary box.

Theorem 1.43 (Orientation compatibility from atlas data).
The explicit positive-Jacobian condition can be obtained from the chosen oriented atlas or oriented-manifold class.

Theorem 1.44 (Local image is a neighborhood).
The inverse function theorem gives that the boundary chart transition maps a small source neighborhood onto a target neighborhood.

Definition 1.45 (Boundary local inverse data).
Local inverse data packages a target boundary box and a right inverse into the source boundary box.

Theorem 1.46 (Selected source box gives target inverse data).
A selected source boundary box, an interior boundary point, and oriented atlas data produce the target box and local inverse data needed for chart change.

Definition 1.47 (Boundary selected box image data).
Image data is the version of local inverse information consumed by the integral invariance theorems.

Theorem 1.48 (Selected boundary box image equality).
The boundary chart transition maps the source boundary face exactly to the chosen target boundary box.

Theorem 1.49 (Boundary change of variables).
Mathlib change of variables converts the source boundary integral into the target chart boundary integral, using orientation to remove the absolute value from the Jacobian.

Theorem 1.50 (Outward-first boundary chart integral invariance).
Boundary chart integrals are independent of the selected oriented boundary chart on the local box.

Definition 1.51 (Transition-pullback local half-space integrals).
The half-space local theorem is re-indexed for the transition-pullback of a manifold form.

Theorem 1.52 (Transition-pullback local half-space Stokes).
Local half-space Stokes applies to the transition-pullback chart representative of a chartwise smooth manifold form.

Theorem 1.53 (Boundary chart local Stokes).
A selected oriented boundary chart box carries the half-space theorem to a chart-written local Stokes statement with the outward-first boundary term.

1.4 Global Assembly Targets

Definition 1.54 (Interior chart selected box).
Proposed Lean name: `Stokes.interiorChartSelectedBox`.

This should be the interior analogue of `boundaryChartSelectedBox`, with support contained in an ordinary Euclidean box inside a chart domain.

Definition 1.55 (Interior chart extended box).
Proposed Lean name: *`Stokes.interiorChartExtendedBox`*.
It should add the neighborhood `ContDiffOn` data needed to invoke the Euclidean box theorem.

Theorem 1.56 (Interior chart local Stokes).
Proposed Lean name: *`Stokes.interiorChartLocalStokes`*.
If the support is contained in an interior chart box, the boundary contribution of the box is artificial and cancels.

Definition 1.57 (Unified selected chart box).
Proposed Lean name: *`Stokes.SelectedChartBox`*.
This should be a small sum-like wrapper with interior and boundary cases, not a new geometric theory.

Definition 1.58 (Chart-supported form).
Proposed Lean name: *`Stokes.ChartSupportedForm`*.
It records that the support of the relevant chart representative is contained in one selected chart box.

Theorem 1.59 (Chart-supported interior Stokes).
Proposed Lean name: *`Stokes.chartSupportedInteriorStokes`*.
The interior case of a chart-supported form has zero boundary term.

Theorem 1.60 (Chart-supported boundary Stokes).
Proposed Lean name: *`Stokes.chartSupportedBoundaryStokes`*.
The boundary case is exactly the existing local boundary chart theorem with the auxiliary chart choices hidden.

Theorem 1.61 (Unified chart-supported Stokes).
Proposed Lean name: *`Stokes.chartSupportedStokes`*.
A form supported in one selected chart box satisfies the local manifold Stokes identity.

Definition 1.62 (Selected chart box cover).
Proposed Lean name: *`Stokes.SelectedChartBoxCover`*.
This finite structure should contain selected boxes covering a compact support, separated into interior and boundary boxes.

Theorem 1.63 (Compact support has a finite selected box cover).
Proposed Lean name: *`Stokes.exists_selectedChartBoxCover`*.
This is the main compactness and chart-box selection theorem for global assembly.

Definition 1.64 (Partition of unity subordinate to selected boxes).
Proposed Lean name: *`Stokes.SelectedBoxPartitionOfUnity`*.
It should use mathlib's smooth partition-of-unity API and expose support containment in selected boxes.

Lemma 1.65 (Partition piece support).
Proposed Lean name: *`Stokes.partitionPiece_chartSupported`*.

Each partition-localized form is chart-supported in its selected box.

Definition 1.66 (Partition-localized form)**.**
Proposed Lean name: *`Stokes.partitionLocalizedForm`*.
This packages multiplication of a manifold form by a smooth scalar partition function.

Lemma 1.67 (Exterior derivative of a finite sum)**.**
Proposed Lean name: *`Stokes.extDeriv_sum`*.
The exterior derivative commutes with the finite sum of localized forms.

Lemma 1.68 (Leibniz rule for partition pieces)**.**
Proposed Lean name: *`Stokes.extDeriv_partitionPiece`*.
This is the local calculation identifying the derivative of each scalar-multiplied piece.

Lemma 1.69 (Partition sum recovers the original form)**.**
Proposed Lean name: *`Stokes.sum_partitionLocalizedForm`*.
The finite sum of all localized pieces is the original compactly supported form.

Lemma 1.70 (Bulk integral of a finite sum)**.**
Proposed Lean name: *`Stokes.bulkIntegral_sum`*.
The bulk integral is additive over the finite family of partition-localized forms.

Lemma 1.71 (Boundary integral of a finite sum)**.**
Proposed Lean name: *`Stokes.boundaryIntegral_sum`*.
The boundary integral is additive over the same finite family.

Definition 1.72 (Project-local bulk integral)**.**
Proposed Lean name: *`Stokes.projectLocalBulkIntegral`*.
The global bulk integral is defined or compared using selected oriented chart boxes.

Definition 1.73 (Project-local boundary integral)**.**
Proposed Lean name: *`Stokes.projectLocalBoundaryIntegral`*.
The global boundary integral is defined or compared using the boundary chart invariance already proved.

Theorem 1.74 (Local integral definitions are chart independent)**.**
Proposed Lean name: *`Stokes.projectLocalIntegral_chartIndependent`*.
The selected chart used to compute a local contribution does not change the resulting integral.

Theorem 1.75 (Project-local chart-supported Stokes)**.**
Proposed Lean name: *`Stokes.projectLocalChartSupportedStokes`*.
The chart-supported theorem is rewritten into the chosen global integral API.

Theorem 1.76 (Global bulk integral equals sum of local bulk integrals)**.**
Proposed Lean name: *`Stokes.globalBulkIntegral_eq_localSum`*.
The bulk side of the global theorem is reduced to finitely many selected boxes.

Theorem 1.77 (Global boundary integral equals sum of local boundary integrals)**.**
Proposed Lean name: *`Stokes.globalBoundaryIntegral_eq_localSum`*.
The boundary side is reduced to the same finite selected-box family.

Theorem 1.78 (Finite sum of local Stokes identities).
Proposed Lean name: *`Stokes.sum_localChartSupportedStokes`*.
Summing the local identities gives the finite-cover identity.

Theorem 1.79 (Global compact-support Stokes).
Proposed Lean name: *`Stokes.globalStokes_compactSupport`*.
For a compactly supported chartwise smooth form on an oriented manifold with boundary, the global integral of $\mathrm{d}\omega$ *equals the global outward-first boundary integral of* ω.

Theorem 1.80 (Compact manifold Stokes).
Proposed Lean name: *`Stokes.globalStokes_compactManifold`*.
If the manifold is compact, the compact-support theorem specializes to the standard compact oriented smooth manifold with boundary statement.

Theorem 1.81 (Comparison with mathlib integral API).
Proposed Lean name: *`Stokes.compareProjectLocalIntegralToMathlib`*.
If mathlib later exposes a canonical manifold-form integral, this theorem should identify the project-local definition with it.

1.5 Global Stokes Lean Anchor Alignment

This fragment records the current Lean anchors that refine the open nodes in Section 1.4. Each entry is meant to be usable as a small coordination task: instantiate the named Lean package, discharge the listed explicit hypotheses, then replace the corresponding planned blueprint node by the implemented anchor.

Definition 1.82 (Interior selected and extended chart boxes).
Carries blueprint nodes: *`def:interior_chart_selected_box` and `def:interior_chart_extended_box`. The selected-box anchor records* $a \le b$, *the closed coordinate box inside the chart-transition domain, and support of `ManifoldForm.transitionPullbackInChart` inside that box. The extended-box anchor adds an open neighborhood on which the same chart representative is `ContDiffOn`.*

Remaining explicit hypotheses: *concrete lower and upper corners; target and overlap inclusions; support containment for the chart representative being localized; and the neighborhood smoothness statement. The local Euclidean Stokes theorem for these interior boxes is still a separate task.*

Definition 1.83 (Selected box partition of unity).
Carries blueprint node: *`def:partition_subordinate_selected_boxes`. It also supplies the finite activity bridge needed by `lem:partition_piece_support` through `mem_active_of_mem_fintsupport` and `mem_active_of_mem_tsupport`.*

Remaining explicit hypotheses: *a smooth partition* ρ*; a compact control set* K*; a finite active set; the proof that every partition term active on* K *lies in that finite set; selected extended boxes for all active indices; and, for existence, the compactness argument selecting boxes from chart sources. The current package is interior-chart shaped; adding boundary-box indices or a unified interior/boundary sum wrapper remains open.*

Definition 1.84 (Partition-localized form data).
Carries blueprint nodes: *`def:partition_localized_form`, `lem:partition_piece_support`, and the interior part of `def:chart_supported_form`.*

The anchor stores the scalar coefficient, the localized form, the equality with pointwise scalar multiplication, and the selected-box support data needed to pass a localized form to interior chart-box APIs.

Remaining explicit hypotheses: *smoothness of the scalar partition coefficient; the support proof for $\rho_i\omega$ in the selected box; compatibility with boundary selected boxes; the finite-sum reconstruction of ω; and the exterior derivative finite-sum and Leibniz rules `lem:extderiv_finite_sum` and `lem:partition_piece_extderiv`.*

Definition 1.85 (Boundary orientation map bridge).
Carries blueprint nodes: *the orientation-map content behind `def:boundary_chart_oriented_atlas` and `thm:boundary_orientation_from_atlas`; it is also the clean bridge into `thm:boundary_transition_open_image`, `thm:boundary_change_of_variables`, and `thm:boundary_chart_integral_invariance`.*

Remaining explicit hypotheses: *a tangential linear equivalence for each boundary-coordinate point; equality of its underlying linear map with `boundaryChartTransitionTangentMap`; either a positive determinant or an `Orientation.map` equality; and setwise data upgrading the pointwise bridge to the selected boundary face. Local inverse data and image equality for boundary boxes are still supplied by the boundary-chart image-data packages.*

Definition 1.86 (Project-local integral wrappers).
Carries blueprint nodes: *`def:project_local_bulk_integral`, `def:project_local_boundary_integral`, and the boundary-chart part of `thm:project_local_chart_supported_stokes`. The wrappers fix the project-facing names for the local bulk and outward-first boundary terms.*

Remaining explicit hypotheses: *selected or extended boundary-chart boxes; oriented-atlas or oriented-manifold assumptions when transporting the boundary term to a target chart; and boundary image data identifying the source lower face with the target boundary box. Interior project-local wrappers have not yet been added.*

Definition 1.87 (Finite-sum global assembly data).
Carries blueprint nodes: *`thm:sum_local_stokes`, `thm:global_bulk_eq_local_sum`, and `thm:global_boundary_eq_local_sum`. This is the algebraic finite-cover layer: local Stokes identities for interior and boundary pieces are summed, artificial interior boundary terms are cancelled, and the result is compared with the stored global bulk and boundary real numbers.*

Remaining explicit hypotheses: *the finite active chart set; interior and boundary piece finsets; bulk and boundary real-valued term functions; local Stokes identities for every active piece; reconstruction of the global bulk and boundary integrals from those sums; and cancellation of artificial interior faces. The field `chartChangeInvariance` is presently only recorded as a proposition; the newer final package below makes the chart-change sum equality explicit.*

Definition 1.88 (Final global Stokes data package).
Carries blueprint node: *`thm:global_compact_support_stokes` as the current final Lean theorem anchor. It refines the older assembly package by naming a separate boundary partition sum and by making chart-change compatibility an equality between boundary-chart terms and boundary-partition terms.*

Remaining explicit hypotheses: *all fields of `GlobalStokesData`: finite chart and piece data; local Stokes identities for interior and boundary pieces; global bulk reconstruction; artificial interior-boundary cancellation; chart-change cancellation into boundary partition* *terms; and global boundary reconstruction. Compact support, manifold-with-boundary assumptions, partition-of-unity construction, and actual manifold integral definitions are not in the theorem statement yet; they must be used to build an instance of this data package.*

Definition 1.89 (Project-local final Stokes package)**.**
Carries blueprint nodes: *`thm:project_local_chart_supported_stokes`, `thm:sum_local_stokes`, and the project-local specialization of `thm:global_compact_support_stokes`. This is the most direct finite-sum target for boundary-chart boxes because the local terms are exactly `projectLocalBulkIntegral` and `projectLocalBoundaryIntegral`.*

Remaining explicit hypotheses: *active chart and piece finsets; source and target chart choices for every piece; lower and upper box corners; the equality reducing the represented global bulk integral to project-local bulk sums; pointwise project-local Stokes for every active piece; chart-change cancellation into boundary partition terms; and boundary partition reconstruction. A useful next collaboration task is to instantiate `localProjectStokes` from `projectLocalStokes_of_boundaryChartExtendedBox` or the oriented image-data variants.*

Theorem 1.90 (Compact and mathlib-facing final statements)**.**
Carries blueprint nodes conditionally: *`thm:compact_manifold_stokes` and `thm:compare_mathlib_integral_api` should not be marked closed by the bare algebraic theorem. Instead, they should point to `Stokes.globalStokes` as the final assembly target once the data package is produced from compactness, partitions of unity, boundary orientation, and the chosen integral API.*

Remaining explicit hypotheses: *conversion from compact manifold hypotheses to compact support data; a canonical or project-local global integral definition; comparison with any future mathlib manifold-form integral; and a proof that the selected-box partition/localized-form layer supplies every field of either `GlobalStokesData` or `ProjectLocalGlobalStokesData`.*

1.6 M6 Constructor Anchor Split

This fragment refines the global-assembly anchors into constructor-level handoff tasks from the 20-agent M6 pass. Each item names the Lean module and anchors to use, says which explicit hypotheses it removes from downstream packages, and lists the fields that still have to be supplied by geometry, partition, or integral-reconstruction workers.

Definition 1.91 (Finite active compact boxes)**.**
Module: *`Stokes.Global.FiniteActive` and `Stokes.Global.CompactActiveBoxes`.*

Explicit hypotheses discharged: *once a compact control set K and smooth partition ρ are fixed, `FiniteActiveOnCompact.ofCompact` supplies the finite active set and the pointwise `fintsupport` subset facts. Once `CompactActiveBoxData` is built, the per-index lower and upper corners, $a_i \leq b_i$, support containment in $\operatorname{Icc} a_i b_i$, and the selected interior box facts are recovered by accessors. Once `CompactActiveExtendedBoxData` is built, the older `SelectedBoxPartitionOfUnity` fields are obtained without restating them index by index.*

Fields still required: *the compact coordinate support set for each active index; compactness of those coordinate supports; the equality between the selected `CompactCoordinateBoxSelection.K` and that support; containment of the chart representative's `tsupport` in the coordinate support; target-chart and overlap inclusions for every active box; and one open smoothness neighborhood with `ContDiffOn` for every active index.*

Handoff task: *geometry workers should construct `CompactActiveExtendedBoxData`; partition workers should consume only `toSelectedBoxPartitionOfUnity` and avoid reopening* *the box-selection fields unless a target/overlap inclusion fails.*

Definition 1.92 (Localized support and smoothness controls)**.**
Module: *`Stokes.Global.LocalizedSupport`, `Stokes.Global.CoefficientBoxSupport`, and `Stokes.Global.LocalizedSmoothness`.*

Explicit hypotheses discharged: *the raw support calculation for `transitionPullbackInChart` of $\rho_i\omega$ is replaced by `LocalizedSupportControl`; coefficient-box constructors turn compact coordinate-image support data into the exact selected-box support statement; `LocalizedSmoothnessData` packages the open-neighborhood smoothness witness needed to upgrade from selected to extended boxes.*

Fields still required: *the concrete source and target charts; lower and upper box corners; target and overlap inclusions; a coefficient support or topological-support bound for ρ_i; smoothness of the localized chart representative, either directly or via chartwise smoothness plus scalar smoothness; and a proof that the coefficient being used is the selected partition coefficient.*

Handoff task: *use this layer before building localized interior pieces. It is the right place to solve support and `ContDiffOn` obligations; downstream global constructors should receive a finished support-control or extended-box package, not the raw analytic lemmas.*

Definition 1.93 (Localized interior pieces)**.**
Module: *`Stokes.Global.LocalizedInteriorPieces`.*

Explicit hypotheses discharged: *for one active index, `LocalizedInteriorPiece` turns the coefficient ρ_i, support control, and smooth-neighborhood witness into the localized form, selected box, extended box, `InteriorLocalStokesData`, and project-local interior Stokes equality. For a finite family, `LocalizedInteriorPieces.localProjectEquality` removes the explicit "prove local Stokes for each active interior piece" hypothesis, and `sum_projectInterior_eq_artificialBoundary` removes the finite-sum local-Stokes algebra.*

Fields still required: *the active finset; the coefficient family; for each index, source chart, target chart, lower and upper corners, a `LocalizedSupportControl`, and a smooth-neighborhood witness. This package does not prove that the active finset came from a compact partition, that the finite localized sum reconstructs ω, or that artificial faces cancel.*

Handoff task: *interior workers should output `LocalizedInteriorPieces` plus a map from its artificial boundary term to the chosen face-sum model. The global worker can then combine it with `ArtificialFacePairingData`.*

Definition 1.94 (Partition sum-one reconstruction)**.**
Module: *`Stokes.Global.PartitionSumOne`.*

Explicit hypotheses discharged: *the ad hoc hypothesis*

$$\forall x \in K, \quad \sum_{i\in s} \rho_i(x) = 1$$

is now obtained from `FiniteActiveOnCompact` or `SelectedBoxPartitionOfUnity`. The pointwise reconstruction of the localized finite form sum on K is also available directly, so reconstruction workers do not need to manipulate mathlib's `finsupport` and `fintsupport` APIs.

Fields still required: *the compact set K must actually contain the support region on which the global integral is reconstructed; the coefficient family in later constructors must be definitionally or propositionally the stored partition; and any theorem needing equality outside K must either enlarge K or supply an everywhere coefficient-sum hypothesis separately.*

Handoff task: *use this anchor to fill the* `lem:partition_sum_form` *side of the blueprint. For exterior-derivative reconstruction, pass the resulting coefficient-sum fact to the* `ExtDerivPartitionReconstructionData` *constructors only after checking the statement is on the same domain.*

Definition 1.95 (Artificial face pairing)**.**
Module: `Stokes.Global.ArtificialFacePairing`.

Explicit hypotheses discharged: *the single final-field proof*

$$\sum_x \sum_p \mathrm{interiorBoundaryTerm}(x, p) = 0$$

can now be supplied by a face-level involution. The module flattens the chart/piece/face indices, proves that the nested sum equals the flattened sum, and converts the pairing package into `ArtificialBoundaryCancellationData` *or directly into the* `GlobalStokesData.interiorBoundaryCancellation` *shape.*

Fields still required: *active charts; interior pieces per chart; face labels per interior piece; the signed face-term function; a set-preserving pairing on flattened face indices; the involution proof; opposite-term cancellation for paired faces; zero fixed-point terms; and, when an external per-piece artificial boundary term is already chosen, an equality identifying it with the sum of recorded face terms on active pieces.*

Handoff task: *local-geometry workers should provide the pairing and sign equations. Assembly workers should consume only* `toArtificialBoundaryCancellationData` *or* `interiorBoundaryCancellation_of_facePairing`*; they should not expand the flattened sigma index set unless debugging signs.*

Definition 1.96 (Boundary chart-change pieces)**.**
Module: `Stokes.Global.BoundaryChartChangePieces`.

Explicit hypotheses discharged: *the finite-sum field* `ProjectLocalGlobalStokesData.chartChangeCancellation` *no longer has to be proved by hand. A per-piece oriented change-of-variables package plus a target selected or extended boundary box gives pointwise equality from the source project-local boundary term to the chosen boundary-partition term; the family wrapper then sums those equalities in the exact final-field shape.*

Fields still required: *for every active boundary piece: target boundary chart, target lower and upper corners,* `boundaryChartOrientedChangeOfVariables`*, and either a selected or extended target boundary box. At the family level, one still needs the equality identifying* `D.boundaryPartitionTerm` *with the transported target project-local boundary integral, plus the* `IsManifold I 1 M` *instance required by the chart-change wrappers.*

Handoff task: *boundary workers should build a selected-family or extended-family package immediately after constructing image data. Global workers can then fill* `chartChangeCancellation` *by one theorem call.*

Definition 1.97 (Boundary-piece global constructors)**.**
Module: `Stokes.Global.BoundaryPieces`.

Explicit hypotheses discharged: `BoundaryProjectLocalPieces` *derives local project Stokes from stored extended boundary-chart boxes and instantiates* `ProjectLocalGlobalStokesData`*. The oriented variant uses source extended boxes, target selected boxes, and image data to instantiate* `GlobalStokesData` *directly with no interior pieces.*

Fields still required: *active charts and local pieces; source and target chart choices; source box corners; extended source boundary boxes; the global bulk and boundary real numbers; bulk reconstruction; boundary-partition reconstruction; and the chart-change cancellation field. The oriented variant additionally needs target chart choices, target corners, target selected boxes, image data, and either oriented-atlas chart membership hypotheses or a* `BoundaryChartOrientedManifold` *instance.*

Handoff task: *use this constructor for boundary-only tests and for validating the boundary side of mixed global Stokes before interior artificial-face cancellation is connected.*

Definition 1.98 (Interior and mixed global constructors)**.**
Module: `Stokes.Global.InteriorGlobalConstructor` *and* `Stokes.Global.MixedGlobalConstructor`*.*

Explicit hypotheses discharged: *the interior-only constructor turns stored* `InteriorLocalStokesData` *packages plus artificial-boundary cancellation into a final* `GlobalStokesData` *instance with empty genuine boundary pieces. The mixed constructor separates the final theorem into three handoff outputs: reconstruction data, an interior local-Stokes package, and a boundary local-Stokes package. Its* `toGlobalStokesData` *definition performs only record alignment.*

Fields still required: *for the interior-only path: active charts, interior pieces, local Stokes data per piece, global bulk reconstruction, artificial-boundary cancellation, and the proof that the represented boundary integral is zero. For the mixed path: a* `PartitionReconstructionData` *package, interior and boundary boundary-term functions, local Stokes packages for both sides, artificial interior-boundary cancellation, and boundary chart-change cancellation.*

Handoff task: *use the interior-only constructor as a narrow smoke test for* `LocalizedInteriorPieces` *plus* `ArtificialFacePairingData`*. Use* `MixedGlobalStokesData` *as the main parent-agent assembly target.*

Definition 1.99 (Partition and exterior-derivative reconstruction constructors)**.**
Module: `Stokes.Global.Reconstruction` *and* `Stokes.Global.ExtDerivReconstruction`*.*

Explicit hypotheses discharged: `PartitionReconstructionData` *isolates the two global reconstruction equalities and converts them, together with local/cancellation/chart-change inputs, into the final* `GlobalStokesData` *record. The exterior derivative module supplies the finite-sum* `extDeriv` *wrappers and an augmented reconstruction package whose chartwise exterior-derivative equality can be carried as a named field or produced from an everywhere coefficient-sum-one hypothesis.*

Fields still required: *finite active chart labels; interior and boundary piece finsets; local bulk term functions; boundary partition terms; represented global bulk and boundary values; the two reconstruction equalities; local Stokes on all pieces; artificial-face cancellation; chart-change cancellation; and, for the exterior-derivative package, the chartwise equality comparing* `extDeriv` *of the localized finite sum with* `extDeriv` *of the original chart representative.*

Handoff task: *reconstruction workers should build this package before calling any final Stokes theorem. Analytic workers should decide whether the ext-derivative field is proved from* `PartitionSumOne` *or recorded as an explicit chartwise theorem.*

Theorem 1.100 (M6 constructor dependency order)**.**
Explicit hypotheses discharged: *this order discharges no mathematical field by itself; it records which constructor should be used before a parent worker asks for the next final-data equality.*

Fields still required: *the union of the fields listed in the preceding constructor anchors. In practice, missing fields should be assigned to the earliest constructor that names them, not patched into `GlobalStokesData` directly.*

The intended parent-agent integration order is: first build compact active boxes and partition sum-one reconstruction; then construct localized interior pieces from support and smoothness controls; then provide artificial face pairings for their boundary terms; then build boundary chart-change family data for genuine boundary pieces; finally feed the reconstruction, local-Stokes, cancellation, and chart-change packages into `MixedGlobalStokesData` or directly into `GlobalStokesData`.

1.7 M6.4–M7 Parallel Blueprint Fragment

This fragment is the next handoff layer after the M6 constructor split. Its purpose is not to introduce another final theorem package. The constructor layer is already in place. The remaining work is mathematical field production: compact-support geometry, localized smoothness, artificial-face geometry, boundary chart-change identification, and global integral reconstruction.

Each entry below is organized by Lean module, record field, and theorem anchor. The "constructor status" paragraph says what can already be used without additional design. The "explicit field mathematics" paragraph names the field-level facts that are still mathematical work. The "parallel theorem anchors" paragraph is the intended worker queue.

Definition 1.101 (Completed constructor layer)**.**
Lean modules: *the completed constructor modules are `Theorem`, `Reconstruction`, `ReconstructionWrappers`, `IntegralReconstruction`, `ExtDerivReconstruction`, `ExtDerivOnSupport`, `MixedGlobalConstructor`, `MixedSelectedConstructor`, `BoundaryGlobalConstructor`, `BoundaryChartChangePieces`, `ArtificialFacePairing`, `ArtificialFaceGeometry`, and `LocalizedInteriorPieces`, all under `Stokes.Global`.*

Constructor status: *these modules already turn named local packages, cancellation packages, chart-change packages, and reconstruction packages into `GlobalStokesData`, `ProjectLocalGlobalStokesData`, or `MixedGlobalStokesData`. The final theorems `globalStokes`, `projectLocalGlobalStokes`, `mixedGlobalStokes`, and `selectedMixedGlobalStokes` are now bookkeeping theorem calls once their records are filled.*

Fields still mathematical: *do not add another global constructor to hide these fields. Workers should instead prove the data needed for `globalBulkIntegral_eq_localBulkSum`, `globalBoundaryIntegral_eq_boundaryPartitionSum`, `interiorLocalStokes`, `boundaryLocalStokes`, `interiorBoundaryCancellation`, `chartChangeCancellation`, and the support-local `chartwiseExtDeriv_eq_global_on` field.*

Handoff rule: *M6.4 workers should output one of the existing data packages, not a new final wrapper. M7 workers should assemble those outputs through `SelectedMixedGlobalInput` or through `BulkIntegralReconstructionData.toMixedGlobalStokesData`.*

Definition 1.102 (M6.4 active compact support to selected boxes)**.**
Lean modules: *`ChartCompactImage`, `CompactSupportChartBox`, and `CompactActiveBoxes` under `Stokes.Global`.*

Constructor status: *`ChartCompactImage` and `ActiveChartCompactImages` already convert compact manifold control sets into compact coordinate images and selected coordinate* *boxes.* `ChartCompactSupportData` *and* `ActiveChartCompactSupportData` *already carry the* `tsupport` *containment that converts those boxes into* `interiorChartSelectedBox`*.* `CompactActiveBoxData` *and* `CompactActiveExtendedBoxData` *already expose lower corners, upper corners, support containment, target containment, overlap containment, and selected or extended box accessors.*

Explicit field mathematics: *the hard fields are still the compact coordinate support for each active chart, compactness of that support, the* `tsupport_subset_coordSupport` *proof, and the geometric shrinkage facts*

$$\mathrm{Icc}(a_i, b_i) \subseteq (\mathrm{extChartAt}\ \ I\ i).\mathrm{target}, \qquad \mathrm{Icc}(a_i, b_i) \subseteq \mathrm{chartOverlap}(I, i, i).$$

For extended boxes, workers must also supply an open neighborhood containing each box and the corresponding `ContDiffOn` *proof.*

Parallel theorem anchors: *one worker can prove compact coordinate-image selection from a compact support set; one can prove target and overlap shrinkage for boxes chosen inside chart domains; one can produce the support-containment theorem for chart representatives; and one can lift selected boxes to extended boxes via localized smoothness. Proposed Lean names:*

Proposed Lean name: `Stokes.activeChartCompactSupportData_of_compactSupport`.

Proposed Lean name: `Stokes.compactActiveBoxData_of_activeChartCompactSupport`.

Proposed Lean name: `Stokes.compactActiveExtendedBoxes_of_smoothness`.

Definition 1.103 (M6.4 partition localization and support control)**.**
Lean modules: `Stokes.Global.FiniteActive`, `Stokes.Global.PartitionSumOne`, `Stokes.Global.LocalizedSupport`, `Stokes.Global.CoefficientBoxSupport`, `Stokes.Global.LocalizedSmoothness`, `Stokes.Global.ReconstructionWrappers`, and `Stokes.Global.ExtDerivOnSupport`.

Constructor status: `FiniteActiveOnCompact` *already supplies the finite active set and the active-support membership lemmas.* `PartitionSumOne` *already turns finite activity into the coefficient-sum-one facts used by* `LocalizedFormReconstructionFields.ofCoeffSumEqOneOn`*.* `LocalizedSupportControl` *already converts coefficient support in a selected box into selected-box support for the localized form $\rho_i\omega$.* `LocalizedSmoothnessData` *already packages the open-neighborhood smoothness field needed by extended boxes.*

Explicit field mathematics: *workers still have to prove that the coefficient family used in the reconstruction is exactly the selected partition coefficient, that each coefficient has the stated compact or boxed support, that $\rho_i\omega$ is smooth in the selected chart neighborhood, and that the localized finite sum reconstructs ω on the support set used by the global integrals. The support-local exterior-derivative field* `ExtDerivOnSupportData.chartwiseExtDeriv_eq_global_on` *is still a theorem, not a constructor obligation that should be guessed.*

Parallel theorem anchors: *a partition worker can prove the coefficient-sum-one and localized-form reconstruction on the compact control set; a support worker can build* `LocalizedSupportControl` *from coefficient support; a smoothness worker can prove* `LocalizedSmoothnessData` *from chartwise smoothness plus scalar smoothness; and an exterior-derivative worker can prove the support-local equality using* `extDeriv_transitionPullbackInChart_localizedFormSum`*. Proposed Lean names:*

Proposed Lean name: `Stokes.selectedPartition_localizedFormReconstruction`.

Proposed Lean name: `Stokes.localizedSupportControl_of_partitionCoeff`.

Proposed Lean name: *`Stokes.extDerivOnSupport_of_partitionSumOne`*.

Definition 1.104 (M6.4 interior local pieces and artificial faces)**.**
Lean modules: *`Stokes.Global.LocalizedInteriorPieces`*, *`Stokes.Global.InteriorLocalStokes`*, *`Stokes.Global.ArtificialFacePairing`*, *`Stokes.Global.ArtificialFaceGeometry`*, and *`Stokes.Global.Cancellation`*.

Constructor status: *one `LocalizedInteriorPiece` already gives a localized form, selected box, extended box, `InteriorLocalStokesData`, and project-local equality. A finite `LocalizedInteriorPieces` family already sums those local equalities. `ArtificialFaceGeometryData` already forgets to `ArtificialFacePairingData` and then to `ArtificialBoundaryCancellationData`.*

Explicit field mathematics: *the remaining mathematical fields are the actual face labels for every interior piece, the map from each local artificial-boundary term to the recorded face sum, the pairing on flattened face indices, proof that the pairing preserves the active finite set, involutivity, equality of paired geometric faces, equality of paired unsigned face integrals, and opposite cubical signs. If the chosen pairing has no fixed points, use `ArtificialFaceGeometryData.ofCoordinateFixedPointFreePairing`; otherwise fixed-point zero terms must be proved explicitly.*

Parallel theorem anchors: *one worker can construct the `LocalizedInteriorPieces` family from support and smoothness packages; one can define the geometric face labels and unsigned terms; one can prove the opposite-face pairing; and one can prove that the artificial boundary term used by the local Stokes package equals the face sum consumed by the cancellation package. Proposed Lean names:*

Proposed Lean name: *`Stokes.localizedInteriorPieces_of_activeBoxes`*.

Proposed Lean name: *`Stokes.artificialFaceGeometryData_of_boxes`*.

Proposed Lean name: *`Stokes.interiorBoundaryTerm_eq_artificialFaceSum`*.

Definition 1.105 (M6.4 boundary local pieces and chart-change fields)**.**
Lean modules: *`BoundaryPieces`*, *`BoundaryChartChangePieces`*, *and `BoundaryGlobalConstructor` under `Stokes.Global`, together with the boundary chart modules `LocalStokes`, `SelectedBoxImageConstructor`, and `BoundaryPieceConvenience`.*

Constructor status: *boundary local Stokes can already be packaged either as `BoundaryProjectLocalPieces` or as `OrientedBoundaryProjectLocalPieces`. Selected-target and extended-target `BoundaryChartChangeFamilyData` already produce the finite `chartChangeCancellation` field. Boundary-only global constructors are already available for smoke tests, but the mixed theorem should use the same chart-change data through `SelectedMixedGlobalInput`.*

Explicit field mathematics: *boundary workers still need target chart choices, target lower and upper corners, target selected or extended boundary boxes, image data identifying source lower faces with target boundary boxes, the oriented change-of- variables package for every active boundary piece, and the pointwise field `boundaryPartitionTerm_eq`. Separately, the boundary reconstruction worker must identify the summed `boundaryPartitionTerm` with the chosen global boundary integral.*

Parallel theorem anchors: *one worker can build source boundary local pieces from selected image data; one can construct target selected boxes and local inverse/image data; one can prove oriented change of variables piecewise; and one can prove the `boundaryPartitionTerm_eq` identification used by the chart-change family. Proposed Lean names:*

Proposed Lean name: *`Stokes.boundaryPieces_of_selectedBoxImageData_family`*.

Proposed Lean name: *Stokes.boundaryChartChangeFamilyData_of_imageData.*

Proposed Lean name: *Stokes.boundaryPartitionTerm_eq_targetBoundary.*

Definition 1.106 (M6.5 global integral and exterior-derivative reconstruction)**.**
Lean modules: *Stokes.Global.IntegralReconstruction*, *Stokes.Global.Reconstruction*, *Stokes.Global.ReconstructionWrappers*, *Stokes.Global.ExtDerivReconstruction*, and *Stokes.Global.ExtDerivOnSupport*.

Constructor status: *bulk reconstruction, boundary reconstruction, and exterior-derivative reconstruction have been split into reusable records. A worker may first prove only* `BulkIntegralReconstructionData`*; another worker may add* `BoundaryPartitionFields`*; another may add* `ExtDerivOnSupportData`*. These records already convert back to* `PartitionReconstructionData` *and to the final* `GlobalStokesData` *or* `MixedGlobalStokesData` *shapes.*

Explicit field mathematics: *the project still needs actual definitions or comparisons for the represented global bulk and boundary integrals. The bulk field must prove finite additivity over localized pieces and the equality with the local project-integral terms. The boundary field must prove finite additivity over the boundary partition and compatibility with the boundary chart-change target terms. The exterior-derivative field must prove that the coordinate representative of* $\mathrm{d}(\sum_i \rho_i\omega)$ *equals the coordinate representative of* $\mathrm{d}\omega$ *on the support on which the bulk integral is reconstructed.*

Parallel theorem anchors: *one worker can implement the bulk integral definition and finite-additivity theorem; one can implement the boundary integral definition and finite-additivity theorem; one can prove the* `extDeriv` *support-local reconstruction; and one can write the comparison wrappers from these definitions to the project-local terms. Proposed Lean names:*

Proposed Lean name: *Stokes.globalBulkIntegral_eq_selectedLocalBulkSum.*

Proposed Lean name: *Stokes.globalBoundaryIntegral_eq_boundaryPartitionSum.*

Proposed Lean name: *Stokes.chartwiseExtDeriv_sum_eq_global_on.*

Theorem 1.107 (M7 selected mixed global assembly)**.**
Lean module: *Stokes.Global.MixedSelectedConstructor.*

Constructor status: `SelectedMixedGlobalInput` *is the M7 parent-agent target. Once its fields are supplied,* `selectedMixedGlobalStokes` *gives the final equality recorded by the reconstruction package. The fields* `selectedPartition_active`, `artificialCancellation_active`, `artificialCancellation_pieces`, `artificialCancellation_term`, `chartChange_active`, `chartChange_pieces`, `chartChange_oldTerm`, *and* `chartChange_newTerm` *are alignment fields; they should be solved by choosing common indices and common term functions, not by new analysis.*

Explicit field mathematics: *the only remaining non-bookkeeping inputs are the selected partition, the reconstruction package, the interior local package, the boundary local package, artificial-boundary cancellation, and chart-change cancellation. These are exactly the outputs of the M6.4 and M6.5 entries above.*

M7 theorem anchors: *the compact-support theorem should instantiate* `SelectedMixedGlobalInput` *from a compact support set, a selected partition of unity, local boxes, reconstruction data, and cancellation data. The compact-manifold theorem should reduce to compact support. A later mathlib-facing theorem should compare the project-local integral API to any canonical manifold-form integral exposed upstream. Proposed Lean names:*

Proposed Lean name: *Stokes.globalStokes_compactSupport_from_input.*

Proposed Lean name: *`Stokes.globalStokes_compactSupport`*.

Proposed Lean name: *`Stokes.globalStokes_compactManifold`*.

Theorem 1.108 (Parallel theorem queue for M6.4–M7)**.**
The next parallel pass should avoid record redesign. The natural independent work units are:

1. *compact support to active selected boxes;*
2. *selected partition coefficient support and sum-one reconstruction;*
3. *localized smoothness and support-local exterior derivative reconstruction;*
4. *localized interior local Stokes packages;*
5. *artificial-face geometry and cancellation;*
6. *boundary selected-box image data and oriented chart change;*
7. *global bulk reconstruction;*
8. *global boundary reconstruction;*
9. *final `SelectedMixedGlobalInput` assembly.*

Items 1–3 can run in parallel with the boundary chart-change work, provided all workers agree on the active chart labels and coefficient family. Items 4 and 5 can run as soon as support and smoothness packages exist. Items 7 and 8 should consume the same local term functions used by items 4–6, so the final M7 assembly is mostly field alignment.

1.8 M7 Proof Obligation Graph

This fragment turns the M7 parent target into a fine-grained proof-obligation graph. Each node is intended to be small enough for an independent Lean task: it names the output package or theorem, records the immediate dependencies, and lists the fields that block the node. The final assembly should not introduce a new global constructor; it should fill `SelectedMixedGlobalInput` from the existing M6.4 and M6.5 packages.

Definition 1.109 (M7 shared index spine)**.**
Proposed Lean name: *`Stokes.M7SharedIndexSpine`*.
This node fixes the common finite active index set, selected partition coefficient family, source chart family, and source selected boxes used by all later M7 packages.

Dependency chain: *compact support and partition choices feed the active finset; the active finset feeds interior pieces, boundary pieces, artificial-face terms, and reconstruction sums.*

Parallel queue: *after this node, the support/smoothness queue, artificial-face queue, boundary chart-change queue, and integral-reconstruction queue can start.*

Blocking fields: *exact equality of active finsets across packages; exact equality of the coefficient family; common term functions for project-local interior terms, artificial-boundary terms, and boundary partition terms.*

Definition 1.110 (M7 selected partition witness)**.**
Proposed Lean name: *`Stokes.m7SelectedPartitionWitness`*.
This node instantiates the selected partition of unity for the active compact support and exposes the coefficient family consumed by localized forms.

Dependency chain: `SelectedBoxPartitionOfUnity` *gives coefficient support, finite activity, and sum-one fields; those become the inputs for localized support and localized reconstruction.*

Parallel queue: *coefficient support and sum-one reconstruction can be proved independently once the same active finset and coefficient function have been fixed.*

Blocking fields: *support containment of each coefficient in the selected box; finite active membership lemmas; pointwise sum-one on the compact control set.*

Definition 1.111 (M7 compact support to active boxes)**.**
Proposed Lean name: `Stokes.m7ActiveCompactBoxes`.
This node builds the active selected and extended box data used by interior localized pieces.

Dependency chain: *compact coordinate images produce selected boxes; selected boxes plus localized smoothness neighborhoods produce extended boxes.*

Parallel queue: *target-domain shrinkage, overlap shrinkage, compact coordinate support, and smooth-neighborhood construction can be split across workers.*

Blocking fields: `tsupport_subset_coordSupport`*; compactness of coordinate supports; lower and upper corners with $a_i \le b_i$; target and overlap inclusions; one open* `ContDiffOn` *neighborhood for every active index.*

Lemma 1.112 (M7 localized support controls)**.**
Proposed Lean name: `Stokes.m7LocalizedSupportControls`.
This node proves that every partition-localized chart representative is supported in its selected active box.

Dependency chain: *coefficient support and original-form support combine to bound the support of $\rho_i\omega$; the bound is converted into* `LocalizedSupportControl`.

Parallel queue: *support of scalar multiplication and support transfer through chart representatives can be developed independently.*

Blocking fields: *equality between the chosen coefficient and the selected partition coefficient; topological-support or* `tsupport` *containment for scalar multiplication; chart representative support transfer.*

Lemma 1.113 (M7 localized smoothness witnesses)**.**
Proposed Lean name: `Stokes.m7LocalizedSmoothnessWitnesses`.
This node supplies the smooth-neighborhood witness needed to upgrade selected localized boxes to extended boxes.

Dependency chain: *scalar smoothness of ρ_i, chartwise smoothness of ω, and the product rule for localized forms feed* `LocalizedSmoothnessData`.

Parallel queue: *scalar partition smoothness, chartwise transport, and localized product smoothness can be proved as separate lemmas.*

Blocking fields: *open source neighborhood; containment of the selected box in that neighborhood;* `ContDiffOn` *for the localized chart representative.*

Lemma 1.114 (M7 localized exterior derivative on support)**.**
Proposed Lean name: `Stokes.m7ChartwiseExtDerivOnSupport`.
This node proves the support-local equality between the chart representative of the exterior derivative of the localized finite sum and the chart representative of the exterior derivative of the original form.

Dependency chain: *finite sum reconstruction gives equality of forms on the control support;* `extDeriv` *finite-sum and chartwise transport convert it into the field required by* `ExtDerivOnSupportData`.

Parallel queue: *this can run in parallel with artificial-face and boundary chart-change work once the selected partition witness is fixed.*

Blocking fields: *exact support set for the global bulk reconstruction; finite-sum `extDeriv` equality; restriction of the equality to the integration support.*

Definition 1.115 (M7 localized interior piece family)**.**
Proposed Lean name: *`Stokes.m7LocalizedInteriorPieces`.*
This node constructs the finite family of localized interior pieces and the local project-Stokes equality for each piece.

Dependency chain: *support control gives selected boxes; smoothness witnesses give extended boxes; extended boxes feed `InteriorLocalStokesData`; the finite family feeds the artificial-face queue.*

Parallel queue: *individual active indices can be proved independently, then bundled into `LocalizedInteriorPieces`.*

Blocking fields: *selected box accessors; extended box accessors; local integral term equality; alignment between the per-piece local term and the global project-local term.*

Definition 1.116 (M7 artificial face labels)**.**
Proposed Lean name: *`Stokes.m7ArtificialFaceLabels`.*
This node defines the flattened artificial-face index type and the signed face term attached to each interior localized box.

Dependency chain: *localized interior boxes determine ordinary cubical faces; boundary faces that lie on the real manifold boundary are excluded from this artificial queue; the remaining faces become cancellation terms.*

Parallel queue: *face indexing and unsigned face-integral comparison can proceed independently of the pairing proof.*

Blocking fields: *face-domain containment; proof that the face is artificial; sign convention for lower and upper faces; equality with the boundary term emitted by the local Stokes package.*

Lemma 1.117 (M7 artificial face pairing)**.**
Proposed Lean name: *`Stokes.m7ArtificialFacePairing`.*
This node proves that artificial faces pair by opposite orientation and equal unsigned integral.

Dependency chain: *face labels produce a pairing candidate; geometric equality of paired faces gives equality of unsigned terms; cubical orientation gives opposite signs.*

Parallel queue: *fixed-point-freeness, geometric face equality, unsigned term equality, and sign opposition can be separate lemmas.*

Blocking fields: *active-finset preservation; involutivity; no fixed points or explicit fixed-point-zero proof; paired-domain equality; paired integrand equality; opposite signed face coefficients.*

Theorem 1.118 (M7 artificial boundary cancellation)**.**
Proposed Lean name: *`Stokes.m7ArtificialBoundaryCancellation`.*
This node produces the cancellation field consumed by the mixed global constructor.

Dependency chain: *the local artificial-boundary term is rewritten as the signed face sum; the pairing theorem cancels that sum; the result is the `interiorBoundaryCancellation` field.*

Parallel queue: *this waits only for labels, local-term identification, and pairing; it does not need global bulk or boundary reconstruction.*

Blocking fields: *`artificialCancellation_active`; `artificialCancellation_pieces`; `artificialCancellation_term`; equality between the local Stokes boundary term and the chosen artificial-face sum.*

Definition 1.119 (M7 boundary source piece family)**.**
Proposed Lean name: *`Stokes.m7BoundarySourcePieces`.*
This node constructs source-side boundary local pieces from selected boundary chart boxes.

Dependency chain: *active boundary indices select source boundary boxes; selected image data invokes boundary chart local Stokes; the resulting source terms feed chart-change cancellation.*

Parallel queue: *source image data can be proved independently for each boundary active index.*

Blocking fields: *boundary-active index set; source lower and upper corners; lower-face image data; outward-first orientation convention; equality with the source boundary partition term.*

Definition 1.120 (M7 boundary target image boxes)**.**
Proposed Lean name: *`Stokes.m7BoundaryTargetImageBoxes`.*
This node chooses target boundary boxes and local inverse data for the source-to-target chart-change comparison.

Dependency chain: *source boundary boxes and chart overlap data determine target boxes; local inverse data identifies the source lower face with the target boundary box.*

Parallel queue: *target box selection, local inverse construction, and target support containment can be split by boundary active index.*

Blocking fields: *target chart choices; target lower and upper corners; target selected or extended boxes; source-to-target image equality; support containment after chart change.*

Lemma 1.121 (M7 oriented boundary chart change pieces)**.**
Proposed Lean name: *`Stokes.m7OrientedBoundaryChartChangePieces`.*
This node proves the oriented change-of-variables equality for every active boundary piece.

Dependency chain: *local inverse data supplies the map; boundary chart integral invariance supplies the integral equality; orientation comparison supplies the sign.*

Parallel queue: *each boundary active index is independent once source and target boxes are fixed.*

Blocking fields: *derivative orientation sign; Jacobian or orientation-map compatibility; target-domain containment; pointwise equality of pulled-back boundary representatives.*

Theorem 1.122 (M7 boundary chart-change cancellation)**.**
Proposed Lean name: *`Stokes.m7BoundaryChartChangeCancellation`.*
This node produces the finite chart-change cancellation package used by the mixed global constructor.

Dependency chain: *oriented per-piece chart-change equalities are summed over the active boundary finset; the source sum is identified with the old boundary term; the target sum is identified with the new boundary partition term.*

Parallel queue: *old-term and new-term identifications can be assigned separately after the per-piece chart-change lemmas exist.*

Blocking fields: *`chartChange_active`; `chartChange_pieces`; `chartChange_oldTerm`; `chartChange_newTerm`; `boundaryPartitionTerm_eq`.*

Theorem 1.123 (M7 global bulk reconstruction)**.**
Proposed Lean name: *`Stokes.m7GlobalBulkReconstruction`*.
This node proves that the represented global bulk integral equals the selected sum of local project-bulk terms.

Dependency chain: *finite additivity over localized forms and partition-sum reconstruction reduce the global bulk side to the active local terms.*

Parallel queue: *this can run in parallel with boundary reconstruction after the shared coefficient family and local term functions are fixed.*

Blocking fields: *definition or comparison theorem for the represented global bulk integral; finite additivity over the active finset; equality between local bulk terms and project-local bulk terms; compatibility with the support-local* `extDeriv` *field.*

Theorem 1.124 (M7 global boundary reconstruction)**.**
Proposed Lean name: *`Stokes.m7GlobalBoundaryReconstruction`*.
This node proves that the represented global boundary integral equals the selected boundary partition sum after chart-change normalization.

Dependency chain: *boundary local pieces give source boundary terms; chart-change cancellation rewrites them to target partition terms; finite additivity identifies the resulting sum with the chosen global boundary integral.*

Parallel queue: *finite additivity and target-term comparison can proceed separately once the target boundary partition term function is fixed.*

Blocking fields: *definition or comparison theorem for the represented global boundary integral; finite additivity over boundary active terms; equality between target chart-change terms and the global boundary partition term.*

Definition 1.125 (M7 reconstruction data fusion)**.**
Proposed Lean name: *`Stokes.m7PartitionReconstructionData`*.
This node fuses bulk reconstruction, boundary reconstruction, and support-local exterior-derivative reconstruction into the reconstruction package expected by `SelectedMixedGlobalInput`.

Dependency chain: *bulk reconstruction supplies the left side; boundary reconstruction supplies the right side; support-local exterior derivative aligns the form being integrated; the wrapper constructors produce* `PartitionReconstructionData` *or* `BulkIntegralReconstructionData` *with boundary fields.*

Parallel queue: *this is a synchronization node, not an analysis node.*

Blocking fields: *identical active finset and coefficient family in all three reconstruction inputs; identical local bulk term function; identical boundary partition term function; exact equality of represented global integral names.*

Theorem 1.126 (M7 selected mixed input fields)**.**
Proposed Lean name: *`Stokes.m7SelectedMixedGlobalInput`*.
This node fills `SelectedMixedGlobalInput` *from the completed queues.*

Dependency chain: *shared indices align the selected partition, interior pieces, artificial cancellation, boundary pieces, chart-change cancellation, and reconstruction package; the existing constructor converts the record to* `MixedGlobalStokesData`.

Parallel queue: *this is the final M7 assembly gate.*

Blocking fields: *the eight alignment fields* `selectedPartition_active`, `artificialCancellation_active`, `artificialCancellation_pieces`, `artificialCancellation_term`, `chartChange_active`, `chartChange_pieces`, *`chartChange_oldTerm`, and `chartChange_newTerm`; all should be definitional or short rewrite goals if upstream queues chose the same names.*

Theorem 1.127 (M7 compact-support theorem from selected input)**.**
Proposed Lean name: *`Stokes.globalStokes_compactSupport_from_selectedInput`.*

This node states the compact-support theorem as an application of `selectedMixedGlobalStokes` to the fully populated selected input.

Dependency chain: *compact support creates the selected input; the selected input gives the mixed global Stokes equality; the theorem exposes the project-facing compact-support statement.*

Parallel queue: *no independent work should remain here except theorem-shape adaptation.*

Blocking fields: *theorem statement must use the same global bulk and boundary integral API as the reconstruction package; compact-support hypotheses must expose the selected partition and compact control set expected by the selected input.*

Theorem 1.128 (M7 compact manifold specialization)**.**
Proposed Lean name: *`Stokes.globalStokes_compactManifold`.*

This node specializes the compact-support theorem to the standard compact oriented smooth manifold-with-boundary statement.

Dependency chain: *compactness of the manifold gives compact support for the form or a compact-support wrapper; the compact-support theorem supplies the final equality.*

Parallel queue: *this should wait until the compact-support theorem shape is stable.*

Blocking fields: *compact-support coercion or wrapper; theorem statement matching the chosen manifold integral API; any comparison theorem to a future canonical mathlib manifold-form integral.*

Proposition 1.129 (M7 critical dependency chain)**.**
The critical path is:

1. *shared indices to selected partition;*
2. *selected partition to active boxes;*
3. *active boxes to support and smoothness;*
4. *support and smoothness to interior local pieces;*
5. *interior local pieces to artificial cancellation;*
6. *artificial cancellation to selected mixed input;*
7. *selected mixed input to compact-support Stokes.*

The boundary chart-change queue and the global reconstruction queue attach to this path at the shared-index and local-term synchronization points. They should not force redesign of the final constructor layer.

Theorem 1.130 (M7 parallel worker queue)**.**
The recommended parallel queue is:

1. *Index spine and selected partition: fix active labels, coefficients, source charts, and term names.*

2. *Compact geometry: construct active selected boxes, extended boxes, target inclusions, and overlap inclusions.*
3. *Localized analysis: prove support control, localized smoothness, and support-local exterior derivative equality.*
4. *Interior local queue: build localized interior pieces, artificial face labels, pairing, and cancellation.*
5. *Boundary queue: build source boundary pieces, target image boxes, oriented chart-change pieces, and chart-change cancellation.*
6. *Reconstruction queue: prove global bulk reconstruction, global boundary reconstruction, and fusion into the reconstruction data package.*
7. *Final assembly: fill `SelectedMixedGlobalInput`, call `selectedMixedGlobalStokes`, and expose the compact-support and compact-manifold theorem shapes.*

Queues 2, 3, 4, 5, and 6 can overlap after Queue 1 fixes names. Queue 7 should be mostly record-field alignment.

Remark 1.131 (M7 blocker ledger)**.**
The blockers that should be reported explicitly by future workers are:

1. *active finset mismatch;*
2. *coefficient family mismatch;*
3. *selected or extended box containment not strong enough;*
4. *unavailable `ContDiffOn` neighborhood;*
5. *artificial face term not matching the local Stokes boundary term;*
6. *missing fixed-point-free pairing or fixed-point-zero proof;*
7. *boundary target image data missing;*
8. *orientation sign mismatch in boundary chart change;*
9. *global bulk or boundary integral API mismatch;*
10. *support-local `extDeriv` equality not stated on the integration support.*

A worker blocked by one of these should return the exact field name and the upstream node above that owns it.

1.9 M8 Measure Localization Route

This fragment records the M8 route from the current represented-integral global theorem to a measure-facing Stokes statement. The goal is to make the remaining analytic handoff explicit: indicators localize finite sums, the bulk integrand is identified almost everywhere with the localized exterior derivative integrand, the local scalar integrands are integrable, the boundary measure is reconstructed from selected boundary pieces, and the M8 statement packages those facts without redesigning the M7 constructor layer.

Remark 1.132 (M8 graph conventions).
Figure 1.5 separates three statuses that were previously easy to conflate, and Figure 1.6 zooms in on the next true-proof wave. Green nodes are existing Lean anchors or constructor packages. Orange nodes are not vague future work: each one names a specific mathematical instance that still has to be produced for the existing fields. Blue nodes are theorem targets or final wrappers that should become short once the orange nodes are discharged.

Theorem 1.133 (M8 indicator localization theorem).
Lean modules: `Stokes.Global.IndicatorSupportLocalization` *and* `Stokes.Global.MeasureIntegralLocalization`.

Role in M8: *this is the pure measure-theory bridge from support control to finite additivity. Termwise support containment gives pointwise and a.e. equality between an unlocalized finite sum and the corresponding sum of indicator terms. The Bochner-integral theorem then turns an a.e. equality*

$$F = \sum_{i \in A} \mathbf{1}_{K_i} f_i$$

into

$$\int F \, d\mu = \sum_{i \in A} \int_{K_i} f_i \, d\mu,$$

provided the localized terms are integrable.

M8 handoff: *later workers should not reprove finite-sum integral algebra inside the bulk or boundary packages. They should instead prove the support containment, measurability, integrability, and a.e. equality hypotheses needed to call these indicator theorems.*

Theorem 1.134 (M8 bulk a.e. equality).
Lean module: `Stokes.Global.BulkIntegrandAE`.

Role in M8: *this node identifies the scalar top-degree bulk integrand for* $\mathrm{d}(\sum_i \rho_i \omega)$ *with the scalar top-degree integrand for* $\mathrm{d}\omega$ *for the chartwise measures used by the bulk integral. The existing package is deliberately measure-parametric: a worker supplies the chartwise measure and proves that its a.e. support lies in the chart-support set where the exterior-derivative reconstruction is valid.*

Explicit field mathematics: *the next bulk-localization theorem still has to combine this a.e. equality with the indicator theorem above. Its concrete output should be a genuine measure integral equality for the represented bulk side, not another global Stokes constructor. Proposed Lean names:*

Proposed Lean name: `Stokes.bulkMeasureIntegral_eq_indicatorLocalBulkSum`.

Proposed Lean name: `Stokes.bulkMeasureLocalizationData_of_bulkIntegrandAE`.

Theorem 1.135 (M8 integrability package).
Lean modules: `Stokes.Global.CompactSupportIntegrability` *and* `Stokes.Global.BoundaryIntegrabilityCompactSupport`.

Role in M8: *the indicator and finite-sum integral theorems require integrability of every selected local term. The compact-support wrappers prove the bulk side from continuity or* `ContDiffOn` *control on compact coordinate boxes. The boundary wrappers prove the lower-zero-face integrability needed for source transition-pullback terms, Jacobian-weighted chart-change terms, and target in-chart boundary terms.*

Explicit field mathematics: *M8 workers should instantiate these wrappers with the exact selected boxes and lower-zero face boxes chosen by M6.4–M7. The important alignment fields* *are the selected active set, the selected partition coefficient, the source and target box records, and the maps from those boxes to the support sets used by the actual measures.*

Theorem 1.136 (M8 boundary measure localization).
Lean modules: `Stokes.Global.BoundaryMeasureLocalization` *and* `Stokes.Global.BoundaryCOVMeasureConstructor`.

Role in M8: *this node fieldizes the genuine boundary measure integral. It takes an ambient boundary measure, an ambient boundary integrand, selected boundary piece sets, selected piece integrands, and the comparison from each piece integral to the recorded boundary partition term. The result is the exact finite boundary partition sum consumed by the existing global reconstruction layer.*

Explicit field mathematics: *the boundary worker must prove the a.e. decomposition of the boundary integrand into selected indicator-localized pieces, termwise integrability, and the equality between every selected piece integral and the transported* `boundaryPartitionTerm`*. Chart-change and COV work should feed this node through term equalities, not by changing the M8 statement shape.*

Theorem 1.137 (M8 measure-level Stokes statement).
Lean module: `Stokes.Global.M8Statement`.

Role in M8: *this is the blueprint-facing statement node. The input keeps the existing M7 mixed-constructor data visible and adds a single measure-localization package containing the genuine bulk and boundary measure integrals. The proof projects the data to* `NaturalGlobalStokesInput` *and then to* `NaturalMeasureStokesInput`*; no new final constructor layer is needed.*

Remaining route: *the statement already exists as bookkeeping. The mathematical work is to instantiate* `M8MeasureLocalizationData`*: prove bulk measure localization from the bulk a.e. equality and integrability package, prove boundary measure localization from the boundary measure node, and align both with the same selected active set, target-image family, artificial-face cancellation, and chart-change cancellation used in M7. Proposed Lean name:*
Proposed Lean name: `Stokes.m8MeasureLocalizationData_of_selectedPartition`.

Theorem 1.138 (M8 measure localization worker queue).
The measure-localization work can be split without touching the final constructor API:

1. *prove the selected-box and selected-face support containments needed by the indicator theorem;*
2. *prove the chartwise bulk a.e. equality for the measures used by the bulk integral;*
3. *instantiate bulk and boundary integrability on every selected box or lower-zero face box;*
4. *prove boundary a.e. decomposition and piece-integral comparisons;*
5. *fill* `M8MeasureLocalizationData` *and call* `m8GlobalStokes`.

The queue deliberately leaves the existing M6.4–M7 records in charge of compact support, local pieces, artificial-face cancellation, and chart-change cancellation.

Proposition 1.139 (Current twenty-agent M8 split).
The current parallel wave should be read as a field-owner map, not as twenty independent final theorems. The useful split is:

1. *Bulk a.e. equality:* `BulkIntegrandAE` *under* `Stokes.Global`.

2. *Indicator support localization:* `IndicatorSupportLocalization` *under* `Stokes.Global`*.*
3. *Bulk localization constructor:* `BulkIntegralLocalizationConstructor` *under* `Stokes.Global`*.*
4. *Boundary measure localization:* `BoundaryMeasureLocalization` *under* `Stokes.Global`*.*
5. *Boundary COV measure constructor:* `BoundaryCOVMeasureConstructor` *under* `Stokes.Global`*.*
6. *Boundary compact-support integrability:* `BoundaryIntegrabilityCompactSupport` *under* `Stokes.Global`*.*
7. *Global integral definitions:* `GlobalIntegralDefinitions` *under* `Stokes.Global`*.*
8. *Natural measure constructor:* `NaturalMeasureConstructor` *under* `Stokes.Global`*.*
9. *M8 statement alignment:* `M8Statement` *under* `Stokes.Global`*.*
10. *Artificial-face field reduction:* `ArtificialFaceFieldReduction` *under* `Stokes.Global`*.*
11. *Pure boundary target-image reduction:* `TargetImageFieldReduction` *under* `Stokes.BoundaryChart`*.*
12. *Target-image to global assembly adapter:* `BoundaryTargetImageToAssembly` *under* `Stokes.Global`*.*
13. *Measure box API:* `MeasureBoxAPI` *under* `Stokes.Global`*.*
14. *Bulk measure localization fields:* `BulkMeasureLocalizationFields` *under* `Stokes.Global`*.*
15. *Measure-localization audit:* `MeasureLocalizationAudit` *under* `Stokes.Global`*.*
16. *Focused build and import-graph audit.*
17. *Remaining* `FIELD` *audit and blocker ledger.*
18. *Blueprint and M8 dependency graph maintenance.*
19. *Field-reduction report:* `reports/stokes-m8-field-reduction.md`*.*
20. *Measure-route blueprint report and printable PDF.*

The parent integration rule is: keep pure boundary-chart modules out of `Stokes.HalfSpace`*, keep global adapters in* `Stokes.Global`*, and only promote modules to aggregators after their focused builds and the no-placeholder scan are clean.*

Proposition 1.140 (Current twelve-agent M8 implementation wave)**.**
The current implementation wave is intentionally smaller than the previous twenty-agent restart. It should focus on discharging true M8 instances rather than adding another layer of wrappers. The twelve owners are:

1. *bulk measure-localization fields from selected chart data;*

2. *actual bulk a.e. integrand equality for the chartwise measure;*
3. *measure-box comparison between measure-local and project-local terms;*
4. *bulk localization constructor feeding `M8MeasureLocalizationData`;*
5. *ambient boundary measure and selected-piece indicator decomposition;*
6. *boundary piece integrability and COV-to-partition term equalities;*
7. *target boundary boxes, image equalities, and local inverse data;*
8. *artificial-face support-zero or overlap-pairing geometry;*
9. *boundary orientation bridge from oriented atlas/manifold data;*
10. *compact-support selected interior and boundary box construction;*
11. *final M8 assembly constructor calling `m8GlobalStokes`;*
12. *blueprint and report synchronization.*

The intended success condition is not twenty or twelve new theorem names; it is a smaller number of fields in `M8MeasureLocalizationData` whose values are produced from compact support, selected charts, measures, and orientation data.

Proposition 1.141 (Current eighteen-agent natural compact-support wave)**.** *Figure 1.5 now includes the current eighteen-agent wave as the explicit bridge from the conditional M8 theorem to a natural compact-support Stokes statement. The owners are:*

1. *audit the shortest constructor route from M8 fields to natural compact-support inputs;*
2. *define the compact-support-facing input record without changing the M8 theorem shape;*
3. *construct finite selected chart boxes from compact support and chart domains;*
4. *project selected partitions to the localized-interior fields used by M8;*
5. *build the compact-support bulk measure package;*
6. *adapt the bulk measure package to `M8MeasureLocalizationData`;*
7. *build the compact-support boundary measure package;*
8. *adapt boundary measure reconstruction to the M8 boundary field;*
9. *construct target-image data from local inverse and compact-image selections;*
10. *adapt target-image data to the global M8 assembly fields;*
11. *feed artificial-face support-zero or pairing data into the M8 artificial-face fields;*
12. *connect the boundary-chart sign convention to the induced oriented boundary orientation;*
13. *align represented global integrals with the canonical compact-support integral interface;*

14. *state the compact-support-facing M8 theorem as a short wrapper around* `m8GlobalStokes`*;*

15. *assemble a natural compact-support input constructor from the pieces above;*

16. *integrate the new modules serially through aggregators after focused checks;*

17. *maintain this blueprint graph and printable PDF without running Lean;*

18. *perform the final parent-side serialized build and no-placeholder scan.*

The mathematical bottlenecks are still Agents 3, 12, and 13: finite chart-box selection from compact support, induced boundary orientation, and the canonical integral comparison. The other owners mostly reduce field surface area by turning manual M8 fields into named constructors.

Proposition 1.142 (Proof-wave true-proof nodes)**.**
The next proof wave should not add another layer of M8 wrappers. Its purpose is to turn the orange mathematical inputs in Figure 1.5 into theorem-produced data. Figure 1.6 isolates this proof wave so that workers can take one node without editing the M8 assembly wrapper. The following nodes are the intended isolated proof tasks.

1. ***Bulk a.e. theorem.*** *Prove the selected-partition version of* `BulkIntegrandAEData`*: for the chartwise bulk measures actually used by the global integral, the scalar integrand of* $\mathrm{d}(\rho_i\omega)$ *agrees a.e. with the localized exterior-derivative integrand on the support box. This is the true proof behind* `ExtDerivToBulkMeasure`*; it should output the a.e. input consumed by* `CompactSupportBulkMeasure`*.*

2. ***MeasureBoxAPI comparison.*** *Prove that the measure-local bulk term for each selected chart is definitionally or theorem-equal to the project-local box term produced by the half-space/local Stokes package. This is the equality that lets the bulk finite sum in* `BulkIntegralLocalizationConstructor` *be the same finite sum that M8 receives from local Stokes.*

3. ***Boundary measure theorem.*** *Instantiate* `BoundaryMeasureLocalizationData` *from the actual boundary measure, selected boundary-piece sets, boundary integrands, integrability on lower zero faces, and the a.e. indicator decomposition of the boundary integrand. The output should be* `boundaryMeasureIntegral_eq_partitionSum`*, not a new statement of Stokes.*

4. ***Artificial support-zero route.*** *For compactly supported localized forms whose support lies strictly inside the selected interior chart box away from artificial faces, prove that every artificial face contribution in the local box theorem is zero. This feeds* `ArtificialFaceSupportZeroToM8`*.*

5. ***Artificial adjacency route.*** *In parallel with the support-zero route, construct the adjacent-face pairing for selected interior boxes: opposite orientation, the same geometric face, and equal unsigned face term. This feeds* `ArtificialFacePairingToM8`*. Only one of the support-zero or adjacency routes is needed for a first natural theorem, but both are legitimate proof targets.*

6. ***Boundary orientation bridge.*** *Connect the project* `BoundaryChartOrientedAtlas` *predicate to the intended mathlib oriented-manifold data and induced outward-normal-first boundary orientation. This should justify the sign used by the lower-zero-face local Stokes theorem across boundary chart changes.*

These six nodes are the current true-proof frontier. The surrounding M8 input builders, compact-support statements, target-image adapters, and canonical-integral skeletons are useful only insofar as they expose these exact proof obligations cleanly.

Remark 1.143 (Focused-build policy for parallel M8 waves)**.**
Parallel Lean workers should avoid running a full `lake build`. Each worker should build only the narrowest target that covers its files, such as `lake build Stokes.Global.BulkIntegrandAE` or `lake build Stokes.BoundaryChart.TargetImageFieldReduction`. The parent integration step should serialize the final full build only after workers are complete and aggregator imports are stable. Documentation-only workers should not run Lean builds at all. This is a build-engine constraint, not a mathematical shortcut: it avoids concurrent `.lake/build` writers and keeps failures local to the responsible owner.

Remark 1.144 (M8 P0/P1/P2 blocker ledger)**.**
The remaining blockers should be reported with priority and field name.
P0: real measure-localization blockers.

1. *`bulkIntegralLocalizes` should be filled through `BulkMeasureLocalizationTermFields.bulkIntegralLocalizes` or the `BulkIntegralLocalizationInput` constructor. The real missing data are the chartwise bulk measure terms, the a.e. integrand replacement for the actual measure, integrability of every localized term, and the measure-box comparison saying that the measure-local term is the project-local box term.*

2. *`boundaryMeasureIntegral` and `boundaryMeasureIntegral_eq_partitionSum` should be filled through `BoundaryMeasureLocalizationData` or the COV constructor. The real missing data are the ambient boundary measure and integrand, the a.e. indicator decomposition over selected boundary pieces, termwise integrability, `globalBoundaryIntegral = boundaryMeasureIntegral`, and piece-integral equality with the selected partition terms.*

P1: geometric instances behind existing constructors.

1. *Artificial faces are fieldized by `ArtificialFaceResolvedData`. The missing instance is either a real pairing of opposite selected artificial faces with equal unsigned terms and opposite signs, or a support-zero proof showing that every localized artificial face term vanishes.*

2. *Target-image selection is fieldized on the boundary-chart side and has a global adapter. The missing instance is the local-inverse/open-map construction producing target selected boxes, image equality, local bijection, and the alignment between target boxes, partition boxes, and selected boxes.*

P2: packaging and API alignment.

1. *Boundary reconstruction wrappers and COV wrappers already exist, but selected representatives still have to use the same term names as the assembly layer.*

2. *Import hygiene remains part of the task: global adapters may import boundary-chart work, but `Stokes.HalfSpace` must not import `Stokes.Global`.*

3. *The final M8 statement should remain a projection from existing M7 data plus* `M8MeasureLocalizationData`*; do not add a new final constructor layer unless a field genuinely cannot be expressed there.*

Theorem 1.145 (M8 to manifold Stokes dependency ladder)**.**
The route from M8 to the usual manifold theorem should be split into the following assignable layers.

1. *Local layer: half-space and boundary-chart local Stokes supply the project-local bulk and outward-first boundary terms.*
2. *Geometry layer: compact support and chart sources select finite interior and boundary boxes, extended boxes, and target image boxes.*
3. *Partition layer: a smooth finite active partition produces localized forms, support containment, smoothness on the chosen neighborhoods, and support-local exterior-derivative equality.*
4. *Measure layer: the bulk and boundary measure-localization packages identify the genuine global integrals with the selected finite sums.*
5. *Cancellation layer: artificial interior faces cancel, and real boundary terms are transported through oriented boundary chart changes.*
6. *M8 assembly layer:* `M8MeasureLocalizationData` *plus the M7 selected mixed input gives* `m8GlobalStokes`*.*
7. *Compact-support statement: the selected partition construction fills the M8 input from a compactly supported chartwise smooth form.*
8. *Compact manifold statement: compactness supplies the compact-support wrapper, yielding the ordinary compact oriented smooth manifold-with-boundary theorem.*
9. *Mathlib comparison: if mathlib later provides a canonical manifold differential-form integral, prove that the project-local integral agrees with it.*

This ladder is the intended division of labor after M8: early workers should discharge orange geometric and measure instances; later workers should only assemble stable records into theorem statements.